

Time-Dependent Association Between Breast Cancer and Risk of Ischemic Stroke

A Nationwide Cohort Study

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Abstract

Background and Objectives

The association between breast cancer diagnosis and treatment and the risk of incident ischemic stroke remains unclear. We investigated ischemic stroke risk among breast cancer survivors and evaluated associations by age, follow-up duration, and type of cancer treatment.

Methods

We conducted a nationwide, retrospective, matched cohort study using the Korean National Health Insurance Service database. Women aged 18 years and older with newly diagnosed breast cancer who underwent breast cancer surgery between January 2010 and December 2016 and had no prior stroke were identified. Each was matched 1:3 by birth year to cancer-free women. The primary outcome was first ischemic stroke, defined as hospitalization with International Classification of Disease, Tenth Revision codes I63/I64 plus inpatient brain CT or MRI. Subdistribution hazard ratios (sHRs) and 95% CIs were estimated using Fine-Gray models that accounted for death as a competing risk and adjusted for sociodemographic factors and cardiovascular and non-CV comorbidities.

Results

We analyzed 107,606 breast cancer surgery survivors (mean age, 50.0 years) and 322,818 matched cancer-free women. Over a mean 7.2-year follow-up, ischemic stroke occurred in 1,155 survivors (1.07%). Stroke risk was elevated shortly after breast cancer diagnosis (1-year sHR 1.59; 95% CI 1.34–1.89; 3-year sHR 1.17; 95% CI 1.05–1.30) compared with cancer-free women, with stronger associations at 3 and 6 months after diagnosis across all age groups. Over the long term, survivors had a slightly lower risk of stroke (sHR 0.94; 95% CI 0.88–1.00), and in a 1-year landmark analysis including only event-free individuals, the risk was lower (sHR 0.87, 95% CI 0.81–0.93). Among survivors, anthracycline use (sHR 1.25) and combined tamoxifen-aromatase inhibitor therapy (sHR 1.49) were associated with increased risk of stroke, whereas radiation therapy was associated with decreased risk (sHR 0.84). These associations attenuated and became nonsignificant beyond 1 year. Stroke risk was also higher among survivors with low income, hypertension, diabetes, or current smoking.

Discussion

The association between breast cancer and ischemic stroke risk is time dependent, with a short-term increase after diagnosis and treatment followed by a gradual decline over time. These

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Supplementary Material

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Glossary

BMI = body mass index; **CHD** = coronary heart disease; **CKD** = chronic kidney disease; **CV** = cardiovascular; **ICD-10** = International Classification of Disease, Tenth Revision; **NHIS** = National Health Insurance Service; **PAD** = peripheral artery disease; **sHR** = Subdistribution hazard ratio; **T2D** = type 2 diabetes.

findings highlight the need for proactive stroke risk management, including early CV assessment and ongoing monitoring for thromboembolic events during survivorship.

Introduction

Breast cancer survivors face a substantial burden of long-term health consequences, including an increased risk of cardiovascular (CV) morbidity and mortality.¹ Stroke, the second leading cause of death and disability worldwide,² can significantly affect breast cancer survivors' quality of life and survival. Furthermore, the combined effects of these chronic conditions can place a considerable economic burden on survivors, caregivers, and the health care system.³

Although several mechanisms may link breast cancer and the risk of ischemic stroke (e.g., direct effects of cancer and its treatment) or shared risk factors (e.g., advancing age, obesity, and underlying CV disease), the specific relationship between breast cancer survivorship and stroke risk remains complex and inconclusive.⁴ Epidemiologic studies investigating this association have yielded conflicting results, and previous meta-analyses have reported no significant association.^{4,5} Some studies report an increased stroke risk,⁶⁻¹⁰ whereas others find no significant association¹¹⁻¹³ or even a decreased risk.¹⁴ Moreover, the association may vary depending on cancer treatment modalities.^{10,15}

These discrepancies are likely due to the complex interplay of individual risk factors, cancer treatments, and follow-up duration. Existing research is often limited by narrow age ranges,^{8,12} short follow-up periods,^{8,11} relatively small sample sizes,^{11,14} lack of specific focus on ischemic stroke as an outcome,⁹ inadequate consideration of shared risk factors (e.g., lifestyle factors and obesity),^{6,8,11,12} failure to account for competing risks,^{6,7,10-12,14} and incomplete consideration of cancer treatment effects.^{6-8,13} Furthermore, many studies were based on patients with breast cancer enrolled before 2010.^{6,7,11,12,14} Recent studies have explored the association between individual treatment modalities and stroke risk, but the overall excess risk compared with the general population remains uncertain.^{7,15,16}

To address these knowledge gaps and inform clinical practice, a comprehensive understanding of the risk of ischemic stroke in breast cancer survivors is crucial, particularly regarding time-dependent and age-specific associations. By leveraging a large-scale clinical dataset, we aim to provide a more definitive assessment of the association between breast cancer survivorship and the risk of developing ischemic stroke, with a particular focus on the influence of age at diagnosis and follow-up duration.

Figure 1 Data Flow Diagram

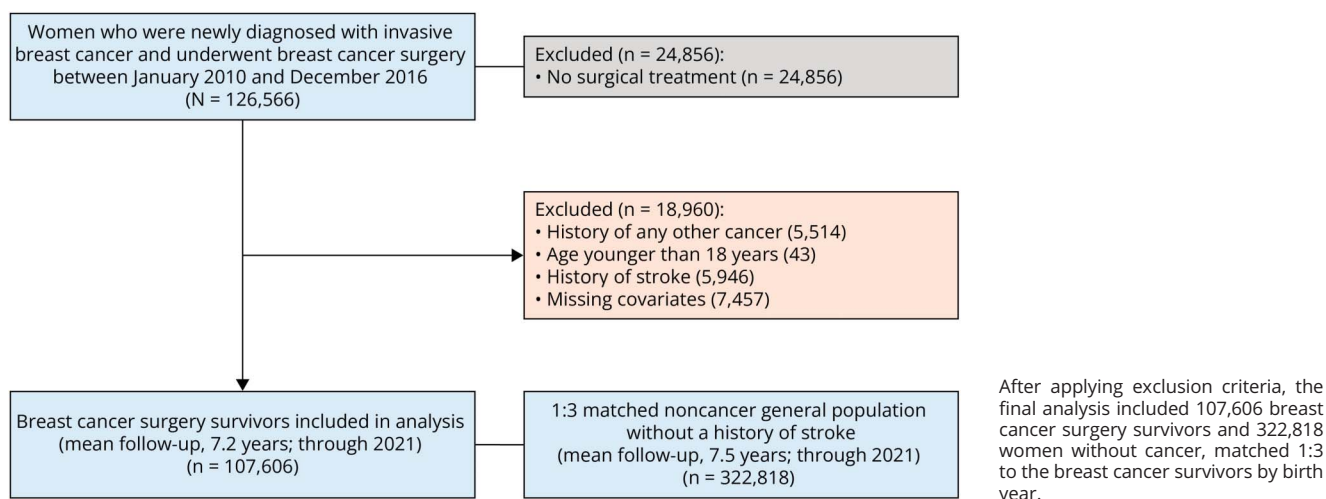


Table 1 Baseline Characteristics of the Study Population and Subgroup of General Health Examination Individuals

Baseline characteristics	Breast cancer n = 107,606	Cancer-free n = 322,818
Age at baseline, mean (SD), y	50.0 (9.4)	50.0 (9.4)
Age group, n (%), y		
18–44	30,815 (28.6)	92,445 (28.6)
45–54	44,815 (41.6)	134,445 (41.6)
55–64	23,916 (22.2)	71,748 (22.2)
≥65	8,060 (7.5)	24,180 (7.5)
Income status, n (%), low	24,122 (22.4)	76,893 (23.8)
Residential location, n (%), urban	54,075 (50.3)	152,986 (47.4)
Comorbidity, n (%)		
Hypertension	18,368 (17.1)	51,048 (15.8)
Type 2 diabetes	5,558 (5.2)	15,601 (4.8)
Hypercholesterolemia	14,316 (13.3)	37,982 (11.8)
Transient ischemic attack	362 (0.34)	1,448 (0.45)
Atrial fibrillation	2,905 (2.7)	5,734 (1.8)
Congestive heart failure	9,594 (8.9)	23,377 (7.2)
Coronary heart disease	17,715 (16.5)	51,852 (16.1)
Peripheral artery disease	1,826 (1.7)	5,746 (1.8)
Cancer treatment modality, n (%)		
Anthracycline	57,967 (53.9)	—
Cyclophosphamide	63,327 (58.9)	—
Taxane	32,027 (29.8)	—
Fluorouracil	15,324 (14.2)	—
Methotrexate or cisplatin	7,221 (6.7)	—
Trastuzumab	15,963 (14.8)	—
Endocrine therapy, n (%)	78,853 (73.3)	—
Tamoxifen only	52,069 (48.4)	—
Aromatase inhibitors only	24,115 (22.4)	—
Both	2,669 (2.4)	—
Radiation therapy, n (%)	76,611 (71.2)	—
Individuals who completed the general health examination		
Age at baseline, mean (SD), y	51.4 (8.7)	51.6 (8.8)
Health behaviors, n (%)		
Current smoker	2,742 (3.9)	6,431 (3.6)
Moderate-to-heavy drinking (≥10 g/d)	17,878 (25.4)	45,567 (25.4)
Regular physical activity ^a	12,901 (18.4)	33,439 (18.7)
Anthropometric and laboratory measurements, mean (SD)		
Height, cm	157.4 (5.5)	156.9 (5.6)

Continued

Table 1 Baseline Characteristics of the Study Population and Subgroup of General Health Examination Individuals
(continued)

Baseline characteristics	Breast cancer n = 107,606	Cancer-free n = 322,818
Body mass index, kg/m ²	23.5 (3.3)	23.5 (3.3)
Obesity (body mass index ≥25 kg/m ²)	20,407 (29.0)	51,713 (28.9)
Waist circumference, cm	76.8 (8.6)	76.8 (8.6)
Fasting glucose, mg/dL	96.7 (20.6)	96.2 (20.6)
Systolic blood pressure, mm Hg	119.8 (15.0)	119.4 (14.9)
Diastolic blood pressure, mm Hg	74.4 (10.0)	74.3 (9.9)
Total cholesterol, mg/dL	199.2 (36.9)	199.5 (37.0)
Chronic kidney disease ^b	2,959 (4.2)	7,511 (4.2)

^a Defined as at least 30 minutes of moderate physical activity for ≥5 d weekly or at least 20 minutes of strenuous physical activity ≥2 d weekly.

^b Defined as either claims-based chronic kidney disease or estimated glomerular filtration rate <60 mL/minute/1.73 m².

Methods

Data Source

This retrospective cohort study used data from the Korean National Health Insurance Service (NHIS) database, a comprehensive and validated resource covering approximately 97% of the South Korean population.^{17,18} With mandatory universal coverage under a single-payer public health insurance system, the NHIS database is highly representative of the South Korean population. The database contains socio-demographic information, detailed claims data, including medical procedures; prescriptions; outpatient visits; hospital admissions; International Classification of Disease, Tenth Revision (ICD-10) diagnostic codes; and treatment details. It also includes biannual health screening data with anthropometric, laboratory, and lifestyle information.¹⁹ This comprehensive structure ensures near-complete capture of clinically attended strokes nationwide and provides a robust source for epidemiologic research.

Women newly diagnosed with invasive breast cancer between January 1, 2010, and December 31, 2016, were identified using ICD-10 C50 and the cancer-specific reimbursement code V193, which is issued to patients registered for payment deductions in South Korea. Eligible patients were women aged 18 years and older with newly diagnosed breast cancer who underwent surgery (herein referred to as breast cancer survivors). Those without breast cancer surgery were excluded because this may indicate advanced or palliative disease. Women with prior cancer or stroke were also excluded. Each breast cancer survivor was matched 1:3 (without replacement) to female NHIS enrollees by exact birth year. Comparators were randomly selected from women alive and insured in the survivor's diagnosis year (index year) with no prior cancer or stroke. The same index year was assigned to align calendar time and follow-up.

Standard Protocol Approvals, Registrations, and Patient Consents

This study was approved by the Institutional Review Board at the Samsung Medical Center (Seoul, South Korea; Samsung Medical Center 2020-03-108). All data used in the analysis were anonymized and deidentified, thus waiving the need for informed consent.

Measurements

The primary outcome was the incidence of ischemic stroke during the follow-up period, identified with ICD-10 codes I63 and I64, and confirmed by claims for either brain MRI or CT during hospitalization. This definition yielded a positive predictive value of 92.2% and sensitivity of 91.2%²⁰ and aligns with those used in previous epidemiologic studies using the Korean NHIS database.^{21,22} Given the low stroke incidence²³ and imaging requirement, specificity is expected to be high.

Detailed information regarding breast cancer treatment within 1 year of diagnosis was extracted from claims data. Chemotherapy included at least 1 cycle of anthracyclines (epirubicin or doxorubicin), cyclophosphamide, and taxane-based regimens (docetaxel or paclitaxel). Targeted therapy involved at least 1 cycle of trastuzumab. Endocrine therapy comprised tamoxifen, aromatase inhibitors (anastrozole, exemestane, or letrozole), or both, determined by the initial prescription. Radiation therapy consisted of at least 1 local or regional treatment.

Comorbid medical conditions present before the index date (breast cancer diagnosis or the corresponding date for the comparison group) were comprehensively assessed using past medical history, clinical and pharmacy ICD-10 codes, and laboratory test results.²⁴ While condition-specific validation metrics are not available for every comorbidity, this

Table 2 Adjusted Subdistribution Hazard Ratios for Developing Ischemic Stroke in Breast Cancer Surgery Survivors Compared to Cancer-Free Women by Age Categories

	Age group		Subjects n	Events n	Observed proportion with stroke (%)	IR per 1,000 PY	Model 1 ^a sHR (95% CI)	Model 2 ^b sHR (95% CI)	Model 3 ^c sHR (95% CI)
Main analysis	All ages	Cancer-free	322,818	3,698	1.15	1.54	1 (ref.)	1 (ref.)	1 (ref.)
		Breast cancer	107,606	1,155	1.07	1.49	0.94 (0.88–1.01)	0.95 (0.89–1.02)	0.94 (0.88–1.00)
	18–44	Cancer-free	92,445	272	0.29	0.39	1 (ref.)	1 (ref.)	1 (ref.)
		Breast cancer	30,815	99	0.32	0.44	1.10 (0.87–1.38)	1.11 (0.88–1.39)	1.04 (0.83–1.31)
	45–54	Cancer-free	134,445	970	0.72	0.97	1 (ref.)	1 (ref.)	1 (ref.)
		Breast cancer	44,815	332	0.74	1.02	1.03 (0.91–1.17)	1.04 (0.92–1.18)	1.01 (0.89–1.15)
	55–64	Cancer-free	71,748	1,248	1.74	2.40	1 (ref.)	1 (ref.)	1 (ref.)
		Breast cancer	23,916	366	1.53	2.19	0.88 (0.79–0.99)	0.89 (0.79–1.00)	0.88 (0.78–0.98)
	≥65	Cancer-free	24,180	1,208	5.00	6.62	1 (ref.)	1 (ref.)	1 (ref.)
		Breast cancer	8,060	358	4.44	6.19	0.90 (0.80–1.01)	0.91 (0.81–1.03)	0.92 (0.81–1.04)
	<i>p</i> Value for interaction						<0.0001	<0.0001	0.0005
During the first year of follow-up	All ages	Cancer-free	322,818	383	0.12	1.19	1 (ref.)	1 (ref.)	1 (ref.)
		Breast cancer	107,606	202	0.19	1.88	1.58 (1.34–1.88)	1.60 (1.35–1.89)	1.59 (1.34–1.89)
	18–44	Cancer-free	92,445	24	0.03	0.26	1 (ref.)	1 (ref.)	1 (ref.)
		Breast cancer	30,815	19	0.06	0.62	2.38 (1.30–4.34)	2.40 (1.31–4.38)	2.28 (1.25–4.16)
	45–54	Cancer-free	134,445	88	0.07	0.65	1 (ref.)	1 (ref.)	1 (ref.)
		Breast cancer	44,815	58	0.13	1.30	1.98 (1.42–2.76)	2.00 (1.43–2.78)	1.96 (1.41–2.73)
	55–64	Cancer-free	71,748	148	0.21	2.06	1 (ref.)	1 (ref.)	1 (ref.)
		Breast cancer	23,916	66	0.28	2.77	1.34 (1.00–1.79)	1.35 (1.01–1.80)	1.34 (1.00–1.79)
	≥65	Cancer-free	24,180	123	0.51	5.10	1 (ref.)	1 (ref.)	1 (ref.)
		Breast cancer	8,060	59	0.73	7.39	1.44 (1.06–1.97)	1.45 (1.07–1.99)	1.49 (1.09–2.03)
	<i>p</i> Value for interaction						0.16	0.16	0.22

Continued

Table 2 Adjusted Subdistribution Hazard Ratios for Developing Ischemic Stroke in Breast Cancer Surgery Survivors Compared to Cancer-Free Women by Age Categories (continued)

	Age group		Subjects n	Events n	Observed proportion with stroke (%)	IR per 1,000 PY	Model 1 ^a sHR (95% CI)	Model 2 ^b sHR (95% CI)	Model 3 ^c sHR (95% CI)
1-Year landmark analysis ^d	All ages	Cancer-free	322,403	3,315	1.03	1.59	1 (ref.)	1 (ref.)	1 (ref.)
		Breast cancer	106,944	953	0.89	1.43	0.87 (0.81–0.93)	0.88 (0.82–0.95)	0.87 (0.81–0.93)
	18–44	Cancer-free	92,420	248	0.27	0.41	1 (ref.)	1 (ref.)	1 (ref.)
		Breast cancer	30,700	80	0.26	0.41	0.95 (0.74–1.23)	0.96 (0.75–1.24)	0.90 (0.70–1.16)
	45–54	Cancer-free	134,351	882	0.66	1.01	1 (ref.)	1 (ref.)	1 (ref.)
		Breast cancer	44,606	274	0.61	0.98	0.94 (0.82–1.08)	0.96 (0.83–1.09)	0.93 (0.81–1.06)
	55–64	Cancer-free	71,593	1,100	1.54	2.45	1 (ref.)	1 (ref.)	1 (ref.)
		Breast cancer	23,726	300	1.26	2.09	0.82 (0.72–0.94)	0.83 (0.73–0.94)	0.82 (0.72–0.93)
	≥65	Cancer-free	24,039	1,085	4.51	6.85	1 (ref.)	1 (ref.)	1 (ref.)
		Breast cancer	7,912	299	3.78	5.99	0.84 (0.74–0.96)	0.86 (0.76–0.98)	0.87 (0.76–0.99)
	<i>p</i> Value for interaction						<0.0001	0.0003	0.01

Abbreviations: IR = incidence rate; PY = person-years; sHR = subdistribution hazard ratio.
^a Crude unadjusted model.
^b Adjusted for age at baseline (continuous variable), household income (4 levels), and residential location (urban, rural).
^c Adjusted for Model 2 + hypertension, type 2 diabetes, hypercholesterolemia, coronary heart disease, congestive heart failure, atrial fibrillation, transient ischemic attack, peripheral artery disease, chronic kidney disease, and medication use including antiplatelet agent, anticoagulant, or statin.
^d The 1-Year landmark analysis includes individuals who remained event-free up to the first year of follow-up.

approach is widely used and accepted in NHIS CV research. These comorbidities included hypertension, type 2 diabetes (T2D),²⁵ hypercholesterolemia, coronary heart disease (CHD), congestive heart failure (CHF), atrial fibrillation (AF),²⁶ TIA, peripheral artery disease (PAD) and chronic kidney disease (CKD) (eMethods for details). For individuals who had undergone a general health examination, the following measures were further considered: hypertension (blood pressure ≥140/90 mm Hg) T2D (fasting glucose level ≥126 mg/dL), hypercholesterolemia (≥240 mg/dL), and chronic kidney disease (the glomerular filtration rate of <60 mL/minute/1.73 m² as estimated by the Modification of Diet in Renal Disease equation).

Income level, a proxy for socioeconomic status, was determined based on monthly health insurance premiums, with the lowest quartile or enrollment in the Medical Aid program defining low-income status.²⁷ Geographic area of residence was categorized as either rural or urban based on the primary local authority districts.

Statistical Analysis

Baseline characteristics were summarized as either means with standard deviations or frequencies with percentages. The follow-up period for each patient extended until the first diagnosis of ischemic stroke, death, or study end (December 31, 2021), whichever came first. The crude incidence rate of ischemic stroke was calculated by dividing the total number of ischemic stroke events by the cumulative person-years of follow-up, expressed per 1,000 person-years. Competing risk survival statistics, accounting for death as a competing risk, were used to estimate the cumulative incidence of ischemic stroke.²⁸ Gray’s test was used to evaluate the potential difference in ischemic stroke incidence between women with and without breast cancer.²⁹ The Fine–Gray proportional subdistribution hazards model was used to calculate the subdistribution hazard ratios (sHRs) and 95% CIs for ischemic stroke incidence, treating death as a competing risk.³⁰ This model was chosen over the cause-specific hazards model to assess the overall effect of breast cancer or its treatment on incident ischemic stroke, rather than to identify specific biological mechanism.³¹ We report

Table 3 Adjusted Subdistribution Hazard Ratios for Developing Ischemic Stroke by Cancer Treatment Type Among Breast Cancer Surgery Survivors

	Main analysis				During the first year of follow-up			1-Year landmark analysis ^a			
	Subjects n	Events n	IR	Adjusted sHR (95% CI) ^b	Events n	IR	Adjusted sHR (95% CI) ^b	Subjects n	Events n	IR	Adjusted sHR (95% CI) ^b
Chemotherapy											
Anthracycline											
No	49,639	547	1.53	1 (ref.)	80	1.62	1 (ref.)	49,395	467	1.51	1 (ref.)
Yes	57,967	608	1.46	1.25 (1.08–1.44)	122	2.11	1.85 (1.29–2.66)	57,549	486	1.35	1.16 (0.98–1.36)
Taxane											
No	75,579	827	1.49	1 (ref.)	140	1.86	1 (ref.)	75,178	687	1.43	1 (ref.)
Yes	32,027	328	1.48	1.05 (0.90–1.22)	62	1.94	0.95 (0.66–1.38)	31,766	266	1.40	1.07 (0.90–1.28)
Targeted therapy											
Trastuzumab											
No	91,643	997	1.50	1 (ref.)	170	1.86	1 (ref.)	91,049	827	1.44	1 (ref.)
Yes	15,963	158	1.42	0.85 (0.71–1.02)	32	2.01	0.88 (0.59–1.32)	15,895	126	1.32	0.84 (0.68–1.02)
Endocrine therapy											
None	28,753	352	1.73	1 (ref.)	70	2.45	1 (ref.)	28,372	282	1.62	1 (ref.)
Tamoxifen	52,069	337	0.87	0.89 (0.76–1.05)	44	0.85	0.60 (0.40–0.89)	51,963	293	0.88	0.96 (0.80–1.14)
Aromatase inhibitor	24,115	425	2.49	0.98 (0.84–1.13)	76	3.16	0.95 (0.68–1.33)	23,967	349	2.38	0.97 (0.83–1.14)
Both	2,669	41	2.50	1.49 (1.06–2.08)	12	4.52	2.13 (1.15–3.95)	2,642	29	2.11	1.33 (0.90–1.96)
Radiation therapy											
No	30,995	444	2.01	1 (ref.)	88	2.86	1 (ref.)	30,637	356	1.87	1 (ref.)
Yes	76,611	711	1.28	0.84 (0.74–0.95)	114	1.49	0.62 (0.47–0.84)	76,307	597	1.25	0.90 (0.78–1.03)

Abbreviations: IR = incidence rate; sHR = subdistribution hazard ratio.

^a The 1-Year landmark analysis includes individuals who remained event-free up to the first year of follow-up.

^b Adjusted for age at baseline (continuous variable), household income (4 levels), and residential location (urban, rural), hypertension, type 2 diabetes, hypercholesterolemia, coronary heart disease, congestive heart failure, atrial fibrillation, transient ischemic attack, peripheral artery disease, chronic kidney disease, medication use (e.g., antiplatelet agent, anticoagulant, or statin), and other cancer treatment modalities (i.e., chemotherapy, targeted, endocrine, and radiation therapies).

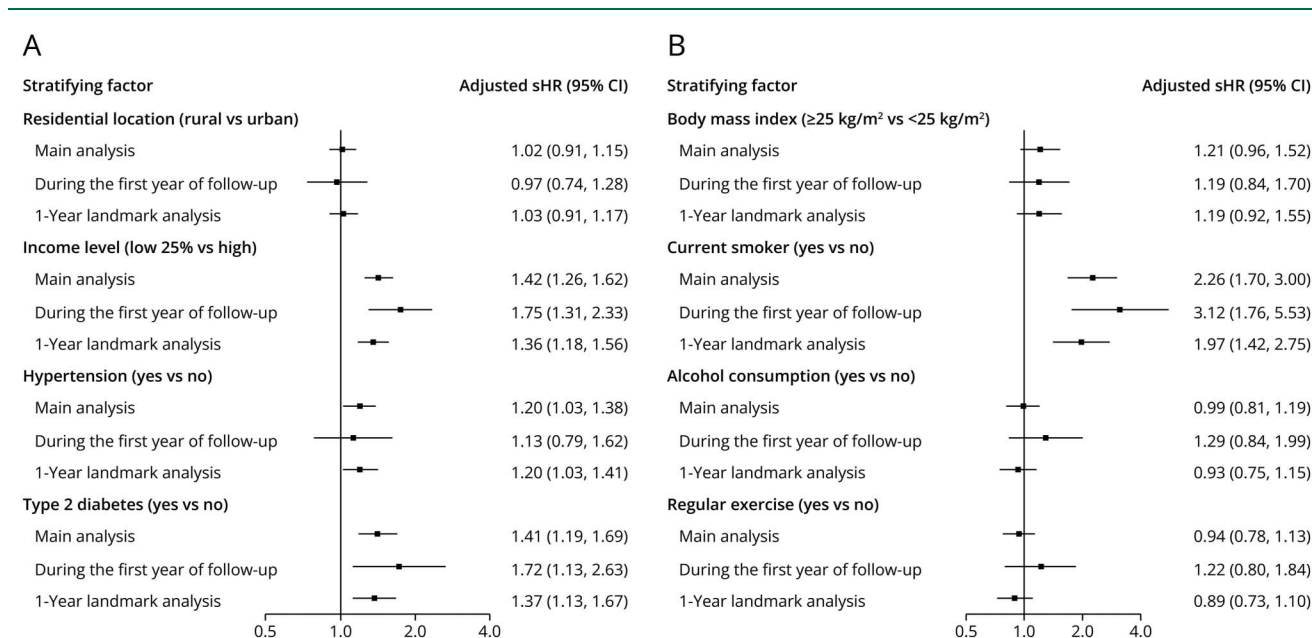
sHRs as relative effect measures. To complement these, we present absolute measures, including the observed proportion with stroke (%) and incidence rates (per 1,000 person-years).

Multiple models were developed to assess the association between breast cancer, its treatment, and the risk of ischemic stroke. Potential confounders were identified a priori based on a literature review. In addition to Model 1, a crude unadjusted model, Model 2 included age at baseline, household income, and residential location to adjust for sociodemographic factors. Model 3 further adjusted for the presence or absence of various comorbidities, including hypertension, T2D, hypercholesterolemia, CHD, CHF, AF, TIA, PAD, CKD, and medication use including antiplatelet agent, anticoagulant, or statin. Among breast cancer survivors, we

further examined ischemic stroke risk associated with different cancer treatment modalities, including anthracyclines, taxanes, trastuzumab, endocrine treatment, and radiation therapies. For example, breast cancer survivors who received anthracycline treatment were compared to those who did not in relation to stroke risk, while accounting for the use of other treatment modalities.

Analyses were stratified by age groups (18–44, 45–54, 55–64, or ≥65 years) to explore potential age-related differences in the associations between breast cancer, its treatment, and ischemic stroke risk. To assess how the risk of ischemic stroke changes over time after a breast cancer diagnosis, we examined the pattern of hazard ratios at specific time points since diagnosis (3 and 6 months, 1, 3, and 5 years) based on the clinical relevance and previous studies.^{7,8,15} In addition,

Figure 2 Adjusted Subdistribution Hazard Ratios for Developing Ischemic Stroke Among Breast Cancer Surgery Survivors Stratified by Selected Factors



(A) Residential location, income level, hypertension, and type 2 diabetes. (B) Body mass index, current smoker, alcohol consumption, and regular exercise among individuals who completed the general health examination. Subdistribution hazard ratios are adjusted for age at baseline (continuous variable), household income (4 levels), residential location (urban, rural), hypertension, type 2 diabetes, hypercholesterolemia, coronary heart disease, congestive heart failure, atrial fibrillation, transient ischemic attack, peripheral artery disease, chronic kidney disease, medication use (e.g., antiplatelet agent, anticoagulant, or statin), and other cancer treatment modalities (i.e., chemotherapy, targeted, endocrine, and radiation therapies) except for the stratifying factor.

landmark analyses at 1, 3, and 5 years post diagnosis were conducted among individuals who were event-free (i.e., no ischemic stroke or death) by each time point. Those who had experienced a stroke or died before the landmark were excluded, and follow-up was reset for the remaining individuals. This approach mitigates immortal time bias and captures how stroke risk evolves during survivorship.³² For example, for the 3-year landmark, the risk set included women alive and stroke-free exactly 3 years after the index date, and follow-up was restarted from that point, counting only strokes occurring thereafter.

Sensitivity analyses were performed with a subset of individuals who had undergone general health screening examinations, including 70,292 women with breast cancer and 179,206 without (65.3% of the main study population), to account for body mass index (BMI), smoking, alcohol consumption, and physical activity (eMethods). In addition, we conducted 1:1 propensity score matching on baseline socioeconomic status and comorbidities in the full NHIS cohort and re-estimated the association between breast cancer and ischemic stroke using Fine–Gray models. The potential for effect modification by income status, residential location, and other comorbidities was examined through stratified analysis and interaction testing with a likelihood ratio test. All statistical analyses were conducted with SAS 9.4 (SAS Institute Inc.). The *p* values provided are two-sided, with the level of significance at 0.05.

Data Availability

The data that support the findings of this study are derived from the Korean NHIS database. These data are not publicly available due to legal and privacy restrictions. Qualified researchers may request access to the NHIS data through formal application and approval by the NHIS. Approved data are accessible only within the NHIS secure analysis center for a limited period, in accordance with NHIS regulations. The study protocol and statistical analysis plan are available from the corresponding author upon reasonable request.

Results

Cohort Selection

A total of 126,566 women with newly diagnosed breast cancer who underwent surgery were initially identified. After sequentially excluding those with any other cancer (*n* = 5,514), those younger than 18 years (*n* = 43), those with prior stroke (*n* = 5,946), and those with missing covariates (*n* = 7,457), the final cohort comprised 107,606 breast cancer surgery survivors. Each survivor was matched 1:3 with 322,818 cancer-free women of the same birth year. Both groups were followed from the index year through December 31, 2020 (Figure 1).

Baseline Characteristics

The average age of the study population was 50.0 years, with 28.6% under the age of 45 (Table 1). Breast cancer survivors

were more likely to have conditions such as hypertension, T2D, hypercholesterolemia, AF, and CHF compared to those without breast cancer.

Among the breast cancer survivors, 53.9% received anthracycline, 14.8% targeted therapy, 73.3% endocrine therapy, and 71.2% radiation therapy. Breast cancer survivors were also more likely to have higher incomes or reside in urban areas. In general, CV risk factors and comorbidities were more prevalent among breast cancer survivors than in individuals without cancer. Baseline characteristics of individuals who had undergone general health examinations were similar to those of the entire study population (eTable 1). There were no substantial differences in smoking, alcohol consumption, regular physical activity, or obesity between women with and without breast cancer.

Risk of Stroke in Breast Cancer Survivors Compared to Women Without Cancer

With a mean follow-up of 7.2 years, 1,155 (1.07%) stroke cases were recorded among breast cancer survivors, compared with 3,698 (1.15%) in the matched cancer-free group. After adjusting for potential confounders, breast cancer survivors had a 6% lower risk of stroke compared to women without cancer (Model 3: sHR 0.94, 95% CI 0.88–1.00), though this result did not reach statistical significance. This risk reduction was more pronounced in the 1-year landmark analysis, where only event-free individuals at 1 year were included (sHR 0.87, 95% CI 0.81–0.93). The inverse association was significant and stronger in women aged 55 and older (*p* value for interaction = 0.01; Table 2). Further landmark analyses at 3- and 5-year follow-up time points indicated a slightly stronger inverse association over time. However, there was no clear trend of decrease in stroke risk over time was observed within each age group (eTable 2).

Cumulative incidence curves (eFigure 1) show higher stroke risk among breast cancer survivors than in the age-matched comparison group during the early follow-up period, with group differences statistically significant up to 3 years (*p* < 0.05) and attenuating thereafter.

However, within the first year of breast cancer diagnosis, stroke risk was 59% higher in breast cancer survivors compared to women without cancer (sHR 1.59, 95% CI 1.34–1.89), with positive associations observed across all age groups. This association was stronger during 3 months and 6 months since breast cancer diagnosis (sHR 2.90, 95% CI 2.17–3.89 for 3 months and sHR 2.27, 95% CI 1.82–2.82 for 6 months). The association risk was attenuated by 3 years post diagnosis (sHR 1.17, 95% CI 1.05–1.30), with breast cancer survivors younger than 45 years at higher risk than other age groups (*p* value for interaction = 0.09). The significantly increased risk was no longer observed overall or within any age group within 5 years postdiagnosis (eTable 2). In contrast, when the analysis was limited to those who had undergone health screening, with additional adjustments for BMI,

smoking, alcohol consumption, and physical activity, the overall association after additional adjustment was null (sHR 0.98, 95% CI 0.90–1.06). At the 1-year landmark, a small but statistically significant lower risk was observed among breast cancer survivors (sHR 0.91, 95% CI 0.83–0.99), with no significant differences by age group (eTable 3).

Risk of Stroke by Treatment Modalities

Those who received anthracyclines had a 25% higher stroke risk compared to those who did not (sHR 1.25, 95% CI 1.08–1.44), with the association being stronger when the analysis was limited to 1 year post diagnosis (sHR 1.85, 95% CI 1.29–2.66). While each endocrine therapy tended to show null associations individually, those treated with both tamoxifen and aromatase inhibitors had an increased risk of stroke compared to those not treated with endocrine therapy (sHR 1.49, 95% CI 1.06–2.08). In contrast, radiation therapy was inversely associated with ischemic stroke (sHR 0.84, 95% CI 0.74–0.95; Table 3). All these associations were stronger 1 year post diagnosis, but were attenuated or became non-significant when analyses were limited to event-free individuals for 1 year. Anthracycline-based chemotherapy was associated with a higher risk of stroke among women aged 55–64 years (sHR 1.28, 95% CI 1.00–1.65), while no significant association was observed in other age groups (eTable 4).

Subgroup and Sensitivity Analyses

In subgroup analyses among breast cancer survivors, those with low-income level (sHR 1.42, 95% CI 1.26–1.62), hypertension (sHR 1.20, 95% CI 1.03–1.38), and T2D (sHR 1.41, 95% CI 1.19–1.69) had increased risk of stroke. Stroke risk was not attenuated in those with hypertension in the 1-year lag analysis. Among breast cancer survivors who had undergone health screening examinations, those who were current smokers had an increased risk of stroke (sHR 2.26, 95% CI 1.70–3.00; Figure 2 and eTable 5). In the 1:1 propensity score-matched cohort, baseline covariates were well balanced after matching (eTable 6), and effect estimates were nearly identical to the primary multivariable estimates (eTable 7).

Discussion

In this large-scale, population-based cohort study with the comprehensive NHIS database, we observed a time-dependent association between breast cancer and ischemic stroke risk. This association is characterized by a short-term increase in stroke risk following breast cancer diagnosis, particularly pronounced within the first year, followed by a long-term decrease in risk. This pattern, observed across age groups, suggests that the impact of breast cancer on stroke risk evolves over time after diagnosis. Notably, the use of anthracyclines and the combined use of tamoxifen and aromatase inhibitors were associated with an increased stroke risk among breast cancer survivors. To our knowledge, this is the largest study examining stroke risk among

breast cancer survivors in the modern era of treatment, considering the complexities of cancer treatments and comorbidities.

Our study aligns with prior research,^{6,7} demonstrating an initial increase in stroke risk, particularly pronounced at 3 months (2.9-fold) and 6 months (2.3-fold) post diagnosis. This early elevation may be attributed to several factors. In addition to cancer's role in accelerating underlying stroke pathophysiology,³³ the invasive diagnostic procedures and surgical interventions involved in breast cancer management can trigger inflammation, endothelial dysfunction, and hypercoagulability, potentially increasing the risk of thromboembolic events and subsequent stroke.³⁴ Postoperative complications may further contribute, as surgery increases venous thromboembolism risk, particularly in patients with cancer due to tumor-related prothrombotic activity.³⁵ In addition, the emotional and physiologic stress associated with a cancer diagnosis and the initiation of treatment can precipitate CV events in susceptible patients, potentially mediated by the release of stress hormones and transient prothrombotic states.³⁶ Allostatic load, which reflects stress burden through cardiometabolic, inflammatory, and immune markers, has also been associated with an increased risk of ischemic stroke in individuals with breast cancer.³⁷ Furthermore, the association between certain cancer treatments (particularly anthracyclines) and stroke risk, supported by our findings, is well documented due to its cardiotoxic effects.^{15,38}

In contrast, the long-term reduction in stroke risk aligns with some previous research suggesting a potential protective effect of breast cancer survivorship.¹⁴ This inverse association might be multifactorial. Increased surveillance and management of CV risk factors in breast cancer survivors, alongside lifestyle modifications post diagnosis, could contribute to this decreased risk.³⁹ In addition, the neuroprotective effects of estrogen⁴⁰ and the selective long-term survival of patients with estrogen receptor-positive breast cancer may also contribute to this reduced risk.⁴¹ The initial higher stroke incidence likely reflects preexisting but undiagnosed CV vulnerabilities. As these patients experience events, the remaining cohort may exhibit a lower overall risk, a phenomenon further supported by our landmark analysis, which indicates a reduced stroke risk in those surviving the initial period following diagnosis. Finally, because death precludes subsequent stroke, higher all-cause mortality among breast cancer survivors compared with the matched comparison group, as accounted for in our Fine-Gray models, could lower the observed cumulative incidence of stroke and contribute to the late attenuation.¹⁰

Among breast cancer survivors, we observed a positive association between anthracycline treatment and the combined use of tamoxifen and aromatase inhibitors and an inverse association of radiation therapies with subsequent stroke risk, which tended to be attenuated and nonsignificant after 1 year. A recent study with the Kaiser Permanente dataset also

showed increased stroke risk with anthracycline treatment, with no distinct pattern found with time since breast cancer diagnosis.¹⁵ While no clear association exists between tamoxifen or aromatase inhibitors alone and stroke risk in breast cancer,⁴² few studies have examined the combined use of these agents. A Swedish study reported the null association of sequential use of tamoxifen and aromatase inhibitors with stroke risk.⁴³ The observed elevated stroke risk with the combined use of tamoxifen and aromatase inhibitors may suggest the potential interactions between these agents and CV health. The inverse association between radiation therapy and stroke risk, particularly in the first year, warrants further exploration, as this may be affected by confounding by indication if women had early-stage breast cancer or fewer comorbidities compared to those receiving more aggressive treatments such as chemotherapy. Radiation therapy does not appear to increase stroke risk in breast cancer,^{44,45} which is consistent with our findings from landmark analysis. The absence of increased stroke risk may reflect the long latency of radiation-induced vasculopathy and the limited carotid exposure in breast cancer radiotherapy.⁴⁶

Our stratified analyses highlighted the increased stroke risk observed in breast cancer survivors with low-income levels. Even though the NHIS in South Korea covers most of cancer treatment costs, reducing the financial strain on these individuals,⁴⁷ this finding emphasizes the importance of addressing the broader issue of socioeconomic disparities in health care access and delivery.⁴⁸ The increased stroke risk in some breast cancer survivors underscores the need for comprehensive strategies to mitigate stroke risk for those with select factors, such as hypertension (optimize blood pressure), T2D (manage glycemia), and current smokers (smoking cessation intervention).^{1,49}

Our study has several strengths, including its large sample size, population-based design, and long follow-up duration, enabling a comprehensive assessment of the time-dependent and age-specific associations between breast cancer and stroke risk. However, our study also has limitations. First, its retrospective design and reliance on claims data may introduce potential biases, including misclassification and incomplete capture of certain clinical variables, such as cancer stage, cardiac imaging, pathology, surgery type, and chemotherapy dosage. While we adjusted for a wide range of potential confounders, residual confounding cannot be entirely ruled out, particularly due to the lack of cancer stage information. The propensity score-matched sensitivity analysis yielded similar results, supporting the robustness of our findings to measured baseline socioeconomic and comorbidity confounding. Second, differential medical surveillance may have influenced our findings, as breast cancer survivors typically receive more frequent follow-up and stricter CV risk-factor management than the general population. These unmeasured care differences, not captured in claims data, may partly explain the lower long-term stroke incidence observed. Third, restricting the cohort to patients who underwent surgery for

breast cancer, may have introduced selection bias, as patients with advanced or metastatic disease who did not receive surgery were excluded, resulting in a study population biased toward earlier-stage, healthier patients. This could underestimate stroke risk in the overall breast cancer population, since the sickest patients (who might have higher stroke risk due to more aggressive disease or treatments) were excluded. In addition to advanced or palliative cases, a small proportion of patients may not have received surgery due to severe comorbidities, death before treatment, or treatment refusal. Nonetheless, approximately 85% of newly diagnosed breast cancer patients in Korea undergo surgery, making our cohort broadly representative of the national breast cancer population.⁵⁰ Fourth, the study population was limited to South Korean women, which may limit generalizability to other settings. Fifth, a small proportion of I64-coded (unspecified) stroke events may have represented hemorrhagic strokes. However, given the requirement for brain imaging and the small number of such cases, any misclassification is likely minimal and nondifferential. Finally, we lacked laboratory markers of hypercoagulability (e.g., D-dimer) that could help clarify stroke mechanisms. Elevated D-dimer is common in cancer patients and has been associated with higher ischemic stroke risk in active cancer.⁵¹ Future studies incorporating D-dimer or other coagulation markers may better delineate the thrombotic contribution to stroke among breast cancer survivors.

This study provides evidence of a complex, time-dependent relationship between breast cancer survivorship and the risk of ischemic stroke, with implications for both clinical practice and public health. The elevated stroke risk shortly after breast cancer diagnosis, particularly in those treated with anthracyclines or the combined use of tamoxifen and aromatase inhibitors, underscores the need for proactive stroke risk management. This includes baseline evaluations before treatment, periodic monitoring during therapy for CV toxicity or thromboembolic events, and optimizing modifiable risk factors such as blood pressure and glucose levels. Future research should clarify these mechanisms and develop strategies to reduce stroke risk in breast cancer survivors.

Authors Contributions

Y.-M.M. Park: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. W. Jung: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. B. Kim: analysis or interpretation of data. M. Lee: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. M.G. Fradley: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. S.J. Malapati: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. T.Z. Naqvi: drafting/revision of the

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